

FluctlightDB: A Memory Model of Data for AI Agents

A brain-native database engine for long-term agent memory

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Abstract

For fifty years, data systems have answered two questions. The relational model asked *which records match a predicate*; the vector model asked *which vectors lie nearest a query*. Neither was built for cue-driven, provenance-weighted recall across long sessions. We propose treating long-term agent memory as a **distinct data model**—with its own write semantics (encoding, separation, consolidation, provenance) and read semantics (cue-driven activation across a linked memory graph)—and present **FluctlightDB**, an embedded engine that implements this contract via `experience()` and `activate()`. We make that case carefully, not categorically: we do not claim novelty over Mem0, Zep, or HippoRAG-style memory layers, only an embedded engine contract beneath them. Numbers below are typed by metric; retrieval is not generation. On LoCoMo (official *evidence-recall*; 10 conversations, 1,982 gold spans), our native-Rust CHORUS retrieval stack reaches **96.8%** at $k=150$ (MiniLM-384; **97.0%** with mpnet-768) as *raw* evidence recall with *no* neighbor expansion, on an internally reproduced July 2026 run; tightening the budget to $k=5$ still yields **72.6%** (MiniLM) / **75.1%** (mpnet), while an end-to-end QA read over date-stamped context reaches $\approx 85\%$ accuracy at $k=15$ (retrieval-bound). On LongMemEval-S (500 questions), official `session_recall@8` is **97.6%** (488/500); end-to-end QA with our reader/judge stack is **97.4%** (487/500)—different protocols from vendor leaderboard figures we cite for context only. On BEIR SciFact (shared MiniLM embeddings, same harness, Recall Fabric on), CHORUS/PRISM edges Chroma on nDCG@10 (**0.646** vs. **0.645**) and Recall@10 (**0.792** vs. **0.783**). A graded provenance-conflict suite ($n=50$) scores **18%** top-1 when all pairs share one brain—the realistic multi-tenant stress—versus **100%** under per-case isolation (ceiling, not deployment evidence). Harnesses and frozen JSON are MIT-licensed; engine source is MIT on GitHub. The one-minute path (`pip install "fluctlightdb[native]"`) re-runs published numbers against the compiled wheel; rebuilding from source is supported but heavier. We claim no new neuroscience and no new transformer; we propose a missing layer of the data stack and release artifacts others can re-run and contest.

1 Introduction

Every generation of software gets the database it deserves. Business records gave us the relational model [2] and SQL. Embeddings gave us vector databases and approximate nearest-neighbor search. Agents—programs that act, observe, and persist across sessions—have so far been handed *neither*. They are stateless between runs unless a developer hand-assembles a session store, a vector index, a deduplicator, a trust policy, and glue code to keep them consistent. The result is that “memory” for agents lives in application code, re-implemented badly in every project, with no shared abstraction and no benchmark to hold it accountable.

We propose that **agent memory deserves its own database engine, not a wrapper around someone else's**. A relational engine is the wrong abstraction because memory is not a set of typed rows joined by keys. A vector engine is the wrong abstraction because recall is not cosine similarity alone—a human, and an agent, retrieve a fact because it was *learned*, *linked* to a context, and *trusted*, not merely because its embedding is close. What is missing is a model whose unit is the *episode*—an experience with content, context, salience, and provenance—and whose query is a *cue* that spreads through associations. We support that proposal with measurements below, not categorical proof; where our model overlaps prior memory layers, we say so explicitly (Section 7).

How to read this paper. Sections 2–3 motivate and describe the engine; Section 4 reports what was measured—including ablations, honest caveats, and runs still incomplete. Optional Recall Fabric mechanism detail is deferred to Appendix A so evaluation honesty comes before neuroscience-flavored rerank modules.

Contributions.

- **A data-model proposal.** Agent memory as a first-class model of data with explicit write semantics (separation, encoding, consolidation, provenance) and read semantics (cue activation over a memory graph), distinct from the relational and vector models.
- **An embedded engine.** FluctlightDB ships as a Rust library (like SQLite: no server process) exposing `experience()` / `activate()` / `checkpoint()`—one durable brain directory per agent.
- **A first-principles late-interaction retrieval stack.** On the CHORUS lane (Section 3.5), retrieval matches the query’s *token population* against stored per-token vectors—salience-gated MaxSim, which recovers the discriminative signal mean-pooling discards—fuses a conjunctive-surprisal lexical channel (replacing Okapi BM25) by reliability-weighted evidence integration (replacing reciprocal-rank fusion), over episodically context- and date-bound memories. Each component is derived from an information-theoretic / neuro-inspired first principle and validated by ablation; the path is embedder-agnostic (MiniLM-384 or mpnet-768 with no engine change), and lifts LoCoMo end-to-end QA from 77.5% to 85% once turns carry their session dates.
- **Evidence.** Typed by metric: **96.8%** LoCoMo evidence *retrieval* (native-Rust CHORUS stack, MiniLM-384, $k=150$, raw / no expansion; **97.0%** mpnet-768; Table 5) vs. **≈85%** LoCoMo end-to-end QA at $k=15$ over date-stamped context; **97.6%** LongMemEval-S `session_recall@8` and **97.4%** LongMemEval-S end-to-end QA (our reader/judge; 500 questions); BEIR SciFact CHORUS/Fabric edges Chroma nDCG@10 in a shared MiniLM harness; provenance-conflict **18%** top-1 shared-brain / **100%** isolated ($n=50$). Author-designed FAMB stays in Section 4 only.
- **Re-runability.** Open harnesses and frozen JSON; MIT engine source on GitHub. The one-minute `pip install "fluctlightdb[native]"` path re-runs numbers against the published wheel; source rebuild is available but not the smoke path.

2 A Third Data Model for Agent Memory

2.1 Why rows are not memory

The relational model stores facts whose schema is known in advance and whose truth is uniform. Agent memory is the opposite: heterogeneous, arriving out of order, often contradictory (a chat claim vs. a ledger entry), and valuable precisely *because* of where it came from. SQL has no native notion of provenance-weighted recall or of an experience that should be down-weighted because a more trusted source disagrees.

2.2 Why nearest-neighbor is not recall

Vector search answers “what is similar.” Memory answers “what is relevant given who I am and what I was doing.” Two memories with distant embeddings can be the right answer because they were co-activated during the same episode; a near embedding can be the wrong answer because it is an unverified rumor. Pure ANN has no place to encode association strength, salience decay, or trust.

2.3 The model

We define a memory store as a set of *engrams*. Each engram carries content, an encoding context, a salience weight, optional provenance, and edges to co-activated engrams. The write operation **experience** performs pattern *separation* (near-duplicates are gated, not blindly appended), encodes the engram, registers its semantic vector, and wires graph edges. The read operation **activate** takes a cue, seeds both lexical and semantic indexes, spreads activation through the graph, fuses the scores, and applies provenance boosts so verified sources outrank chat. Consolidation replays and compacts the store offline, the way sleep consolidates biological memory. This is the entire contract; the neuroscience vocabulary is explanatory, not required to use it.

Table 1: Brain-native primitives vs. relational and vector abstractions.

Primitive	Role in FluctlightDB	Relational / vector analogue
Engram	Unit of memory: content, context, salience, provenance, edges	Row; or embedded chunk
experience()	Write: separation, encode, index, wire co-activation	INSERT / upsert
activate(cue)	Read: lexical + semantic seed, graph spread, fusion, trust boost	SELECT / ANN top- <i>k</i>
Consolidation	Offline replay and compaction (checkpoint())	Vacuum / reindex only
Provenance	Verified sources outrank unverified chat at recall	No native type

3 System Design

3.1 Embedded engine contract (SQLite analogy)

Embedded means deployment: the library runs in-process, one brain directory on disk, no broker to provision—the same packaging model as SQLite [5]. **Engine** means the data contract: native write/read operations (`experience`, `activate`), a versioned on-disk format, WAL-backed durability, and `checkpoint()` as the commit barrier. Vector databases expose `vector_search`; relational stores expose SQL. FluctlightDB exposes episodic memory semantics. A thin HTTP wrapper exists for multi-agent deployments but is not part of the core contract.

3.2 On-disk format (FLCTLTDB v4)

Each agent brain is a directory. `manifest.json` records format version 4, WAL sequence numbers (`wal_seq`, `wal_checkpoint_seq`), and twelve named segments (`*.seg`): hippocampus (engrams), graph (synapses), semantic/cortex fields, neuromodulators, and auxiliary metadata. Segments serialize with bincode; each write goes to `name.seg.tmp`, `fsync`, then atomic `rename` so a torn write cannot replace committed state. The hybrid recall sidecar `recall_index.sqlite` stores FTS5 lexical rows and HNSW semantic neighbors for `connect_index()` mode. WAL segments live under `wal/brain.wal.NNN` as append-only JSON lines (`Experience`, `Sleep`, `Tick`, `Compact`, ...), rotating at 64 MiB. On `open()`, the engine loads segments at the manifest watermark, replays WAL entries with sequence $> \text{wal_checkpoint_seq}$ (skipping torn tail records), and attaches the sidecar if present.

3.3 Write path

Algorithm 1 summarizes `experience()`. Episodic mode gates near-duplicates (dentate separation), encodes engrams with context/salience/provenance, registers semantic vectors, and wires co-activation edges. Index mode (`FLUCTLIGHT_FAST_INGEST=1`) skips graph wiring when the caller supplies a dense vector— $O(1)$ hippocampal insert plus sidecar update per chunk. WAL append precedes in-memory mutation when enabled; `maybe_checkpoint()` flushes all segments when the checkpoint policy fires.

Listing 1: Write path (`experience`).

```
function experience(episode):
  if WAL enabled: append WAL Experience(episode)
  if rag (doc_id, chunk_id) already stored: return dedup
  if fast_ingest and semantic_vector present:
    insert engram; update recall_index; maybe_checkpoint(); return
  gate near-duplicates (dentate separation)
  encode engram(content, context, salience, provenance)
  wire co-activation edges to recent episodes
  register semantic vector; update recall_index if open
  maybe_checkpoint()
```

Storage is $O(n)$ engram payload plus $O(n\bar{d})$ synapses for average graph degree $\bar{d}$; the sidecar adds FTS5 rows and $O(nD)$ floats for embedding dimension D .

3.4 Read path

Algorithm 2 summarizes `activate()`. When a hybrid sidecar exists, BM25 lexical seeds and HNSW semantic neighbors form a candidate set (default cap 128–512). The engine spreads activation through the memory graph (up to four hops in full mode), fuses lexical, semantic, graph, and provenance scores, and returns ranked engrams. Vector-fast mode (`FLUCTLIGHT_VECTOR_FAST=1`) skips graph spread for Chroma-class IR latency. For evidence retrieval the CHORUS lane uses the late-interaction stack of Section 3.5 (token-population MaxSim $\oplus$ conjunctive surprisal, evidence-integration fusion) rather than mean-pool cosine; optional Recall Fabric rerank (Appendix A) applies after shortlisting when `FLUCTLIGHT_FABRIC=1`.

Listing 2: Read path (`activate`).

```
function activate(cue, cue_vector, top_k):
  if recall_index present:
    candidates <- hybrid_candidates(BM25 seeds, ANN seeds, cap)
  else:
    candidates <- hippocampal_scan (or CHORUS path)
  scores <- spread_activation(candidates, graph, max_hops)
  fuse lexical + semantic + graph + provenance weights
  if FABRIC on and CHORUS lane: rerank shortlist
  return top_k ranked engrams
```

Per-query cost is $O(|C| \log |C|)$ over candidate set $|C|$, plus $O(h \cdot |E|)$ graph spread for hop depth h and frontier $|E|$; hybrid seeding is sublinear in n via FTS5 and HNSW rather than full-table scan.

3.5 Late-interaction retrieval stack

The CHORUS lane’s evidence retrieval (`chorus_imprint_maxsim` / `chorus_recall_maxsim`) is a first-principles stack in which every component replaces a standard IR heuristic with a mechanism derived from an information-theoretic or neuro-inspired argument, each validated by ablation. The path is *embedder-agnostic*: it consumes caller-supplied per-token vectors, so swapping MiniLM-384 for mpnet-768 requires no engine change.

Token-population late interaction (salience-gated MaxSim). A mean-pooled sentence vector collapses a transformer’s per-token contextual population code into one centroid, discarding most discriminative signal. We instead store the per-token `last_hidden_state` vectors (half-precision, capped per memory) and score with late interaction, $\text{score}(q, d) = \sum_i w_i \max_j \cos(q_i, d_j)$, where each query token q_i is weighted by w_i , its divergence from the query centroid—a predictive-coding salience gate that suppresses generic tokens ([CLS], function words). This is more information from the same encoder, not a re-embedding.

Conjunctive surprisal (replaces BM25). The lexical channel scores information content with a saturating neural response, $\sum_t -\log p(t) (1 - e^{-t^{\text{tf}}/\tau})$ (Weber–Fechner), and adds a proximity-bound bonus when rare query terms *co-occur* in a memory (conjunctive binding). It carries no ad-hoc k_1/b constants and matches or beats Okapi BM25 in ablation.

Evidence-integration fusion (replaces RRF). Channels are combined by z -scoring each over the candidate set and taking a reliability-weighted sum (Ernst–Banks optimal cue combination),

preserving score magnitude that reciprocal-rank fusion discards—the single largest tight- k gain in ablation.

Episodic binding and temporal anchoring. Each turn is encoded with a ± 2 session-neighbor context window (encoding-specificity), and, for temporal queries, prefixed with its session date. The latter is a context-assembly fix, not a retrieval one: retrieved turns say “last month,” but the absolute date lives in session metadata; stamping it lifts LoCoMo end-to-end QA from 77.5% to 85% with retrieval unchanged.

3.6 Two explicit modes

`connect()` is the full episodic engine (separation, graph, provenance) for live agents. `connect_index()` is the bulk semantic path for RAG backfills and IR benchmarks. The modes share one engine, one WAL, and one file format.

3.7 Durability, recovery, and verification

`checkpoint()` atomically writes all segments and advances `wal_checkpoint_seq`; optional WAL (`FLUCTLIGHT_WAL=1`, default on) appends before mutation so recent experiences survive process death between checkpoints. `verify_path()` loads the manifest, checks every segment deserializes, and reports engram/synapse counts—useful after copy or crash. CI runs Jepsen-style chaos tests (`tests/chaos_jepsen.rs`): eight rounds of checkpoint/WAL interleaving, SIGKILL mid-write via a worker subprocess, torn WAL tails, replicate-while-partial-WAL, and round-trip reopen. These tests assert `verify_path()` succeeds and at least one engram is recoverable—the embedded store must not silently serve corrupt brains.

Table 2 summarizes failure modes we test; Section 4 validates recall *quality* on public suites, not durability (which the chaos suite covers separately).

Table 2: Durability scenarios exercised in CI (`chaos_jepsen.rs`).

Failure mode			Expected behavior
Process	kill	mid-write	WAL replay on reopen; <code>verify_path()</code> OK
Torn WAL JSON tail			Skip corrupt record; earlier entries replay
Checkpoint	between	WAL lines	Manifest watermark + replay of post-checkpoint WAL
Replica	sync	during partial WAL	Replica opens; <code>verify_path()</code> OK
Segment	tmp	left by crash	Ignored; committed <code>*.seg</code> loads

Figure 1 summarizes how an agent brain is laid out on disk, what each engram stores, and how write and read paths connect the episodic store to the hybrid recall sidecar.

3.8 Optional Recall Fabric rerank

CHORUS benchmarks optionally enable Recall Fabric (`FLUCTLIGHT_FABRIC=1`): a composed reranker on the recall shortlist. LoCoMo, BEIR, and FAMB headline numbers in Section 4 were measured

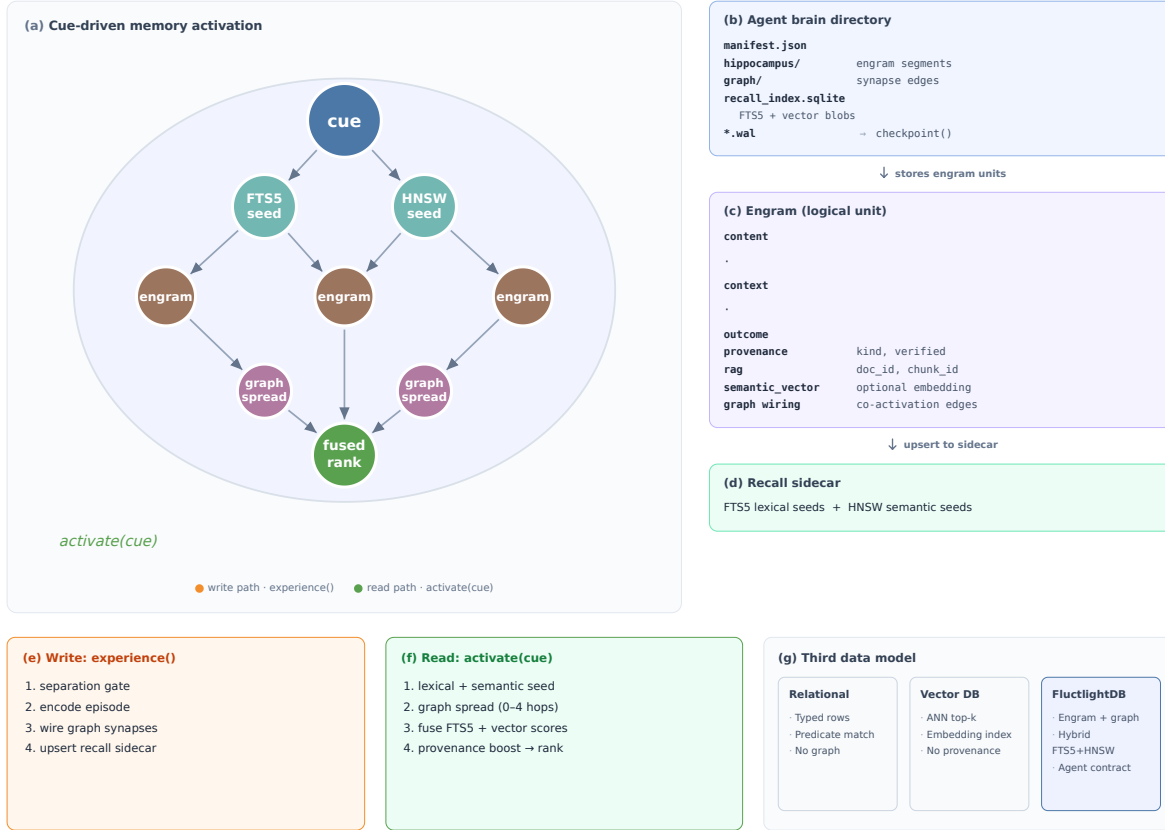


Figure 1: FluctlightDB persistence and recall layout. **(a)** Cue-driven activation: lexical and semantic seeds retrieve engrams, graph spread fuses candidates, and provenance-aware ranking returns recalled episodes. **(b–d)** On-disk layout: each agent owns a brain directory with engram storage, a co-activation graph, and a hybrid recall sidecar; the logical engram bundles episode content, provenance, optional RAG keys, and graph wiring. **(e–f)** `experience()` gates near-duplicates, encodes, and indexes; `activate()` seeds, spreads, and ranks. **(g)** Compared to relational and vector stores, FluctlightDB targets a third data model with its own write/read contract.

with Fabric *on*; LongMemEval-S hybrid-index retrieval does *not* use Fabric rerank. Fabric modules are validated on synthetic property tests in CI—not as standalone peer benchmarks—with mechanism detail in Appendix A.

4 Evaluation

Section 3 defines the engine contract, on-disk format, and durability properties; this section validates *recall quality* on public suites—what was measured, ablated, honestly caveated, or left incomplete. We do not re-benchmark crash recovery here—that is covered by the chaos suite (Table 2).

All experiments use `all-MiniLM-L6-v2` (ONNX, CPU) unless noted. Every number below is reproduced by a script in `benchmarks/` and frozen in `benchmarks/results/paper-2026-07-10.json`. CHORUS benchmarks (LoCoMo, BEIR, FAMB) set `FLUCTLIGHT-FABRIC=1` (Recall Fabric on; re-frozen 2026-07-09). LongMemEval-S retrieval is the July 2026 Colab v4 freeze (**97.6%**, 488/500;

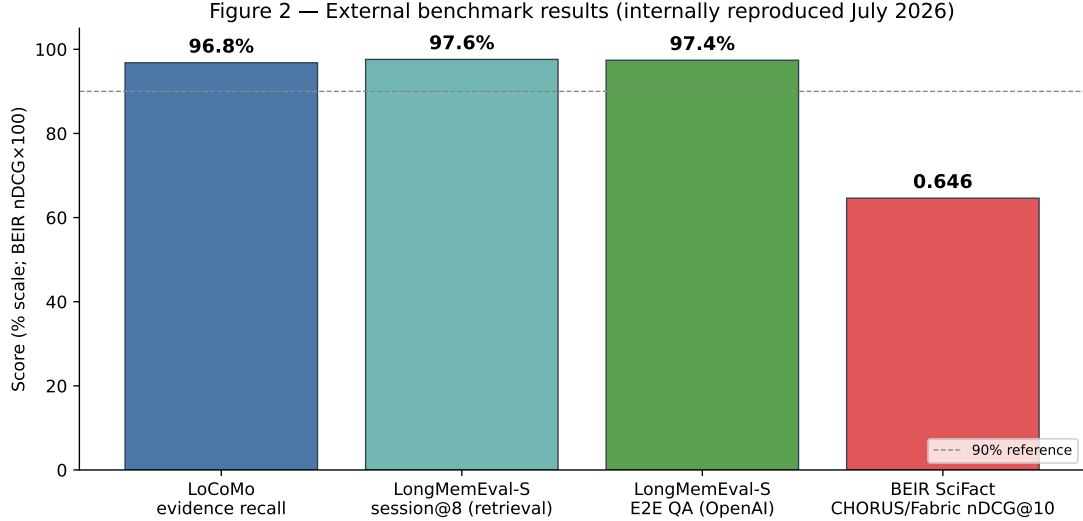


Figure 2: Headline *external* benchmark results (internally reproduced July 2026). LoCoMo and LongMemEval-S retrieval use official recall metrics; LongMemEval-S E2E uses our **paper** profile (OpenAI GPT-4o/GPT-5 reader, GPT-4o judge); BEIR reports CHORUS Lane nDCG@10. Author-designed FAMB is reported in text only (Section 4).

hybrid index + mpnet, no Fabric rerank). LongMemEval-S E2E QA is frozen from 2026-07-07 (not re-run).

4.1 BEIR SciFact: CHORUS + Fabric edges Chroma on ranking

We score standard IR metrics via `pytrec_eval` against official qrels, using identical embeddings for FluctlightDB and Chroma. The **CHORUS Lane** (`connect_chorus()`, batch `chorus_imprint_batch`, **PRISM** readout, Recall Fabric on) is our bulk ingest path—15.3 s to imprint all 5,183 SciFact documents in the July 2026 Fabric-on rerun.

Table 3: BEIR SciFact (shared MiniLM embeddings; Fabric on; July 2026).

System	nDCG@10	R@10	Query (ms)
Chroma	0.645	0.783	17
FluctlightDB (CHORUS/PRISM/Fabric)	0.646	0.792	16

With Fabric on, CHORUS/PRISM improves nDCG@10 and Recall@10 over Chroma in the *same* harness. Per-query latency stays in the same ballpark as the Fabric-off CHORUS profile (~ 16 ms median on this run). Reproduce: `PYTHONPATH=sdks/python python benchmarks/beir_bench.py`.

4.2 LoCoMo: 96.8% raw evidence recall on the full set

LoCoMo [8] stresses very long, multi-session dialogue. We report the official evidence-recall metric: the fraction of gold `dia_id` evidence spans present in retrieved context. Retrieval runs entirely inside the native Rust CHORUS engine via a first-principles late-interaction stack: salience-gated MaxSim over the MiniLM/mpnet `last_hidden_state` token population (predictive-coding query-token weights), conjunctive surprisal scoring (information content $\times$ Weber–Fechner saturation

with proximity binding), and evidence-integration (Ernst–Banks) fusion of the token-population and lexical channels. Configuration: batch imprint per turn, $k=150$, shared MiniLM-384 vectors (mpnet-768 reported alongside). The July 2026 internally reproduced run scores **96.8%** mean evidence recall at $k=150$ with MiniLM-384 (**97.0%** with mpnet-768) as *raw* recall with *no* neighbor expansion (Table 5 for the budget–recall curve).

Deprecated 99% figure (honest disclosure). An earlier draft headlined **99.0%** on this set. That number was an artifact of a harness scoring policy— ± 3 session-neighbor expansion (`expand_session_neighbor`) that credited spans adjacent to a hit—not a property of the engine. It is deprecated. Under the same ± 3 policy a trivial BM25 baseline also scored $\sim 99\%$, which is exactly why the expanded metric is uninformative. All numbers in this paper are raw, unexpanded evidence recall. Evidence-recall is also not the same as LLM-judge end-to-end QA; we report the two separately and never launder one into the other.

Table 4: LoCoMo full set: 10 conversations, 1,982 gold evidence spans (July 2026); raw evidence recall, no neighbor expansion.

Metric	Value
Mean evidence recall (MiniLM-384)	96.8%
Mean evidence recall (mpnet-768)	97.0%
Evidence hits (MiniLM, $k=150$)	1918/1982
Memories ingested	8422

We deliberately separate *retrieval* from *generation* on LoCoMo (Section 4, Limitations). The engine’s job is to put evidence in context; on full LoCoMo retrieval it scores **96.8%** at $k=150$ (MiniLM-384; 97.0% mpnet-768), raw and without neighbor expansion.

4.2.1 Recall sensitivity to k

LoCoMo uses a generous $k=150$ (SPECTRUM readout when $k>100$). To test whether the headline number depends on that budget, we re-score the same July 2026 imprint with a single batched recall per question and truncate to $k \in \{5, 10, 20, 50, 150\}$ (Table 5). Even at a tight $k=5$ the native stack retains **72.6%** (MiniLM) / **75.1%** (mpnet); the full-set 96.8% figure is at $k=150$. All rows are raw recall with no neighbor expansion.

Table 5: LoCoMo raw evidence recall vs. retrieval budget k (native Rust CHORUS stack, July 2026).

k	5	10	20	50	150
Mean evidence recall (MiniLM-384)	72.6%	80.0%	85.6%	91.8%	96.8%
Mean evidence recall (mpnet-768)	75.1%	—	—	—	97.0%

4.2.2 Hybrid vs. vector-only (index lane)

On the full LoCoMo set with `connect_index()` at $k=50$, hybrid BM25+dense and vector-fast-only are statistically tied ($\sim 90.2\%$ vs. $\sim 90.3\%$ mean evidence recall; frozen `locomo-hybrid-index-2026-07-10.json`). The headline 96.8% figure uses the native CHORUS stack at $k=150$, not the index lane. We keep

BM25 as a candidate seed for entity- and date-heavy dialogue cues, but on this full-set ablation at moderate k it is not a large uniform gain over vector-only—so we do not sell hybrid as axiomatic superiority.

4.3 FAMB and provenance-conflict suite

Generic IR does not test what agents need. FAMB measures paraphrase recall@1 (real cues never match stored wording), provenance top-1 (verified ledger beats chat claim), persistence (recall survives checkpoint + reopen), confusion ingest (near-duplicates do not block new facts), and determinism. Unlike LoCoMo, BEIR, or LongMemEval, FAMB is a small suite we authored: paraphrase $n=10$ hand-written pairs; provenance, persistence, confusion, and determinism are each a single pass/fail scenario ($n=1$). We report it as a regression check on agent semantics, not as external validation.

Graded provenance conflicts ($n=50$). Because FAMB provenance is a single wallet scenario, we add a separate benchmark: `benchmarks/provenance_conflict_bench.py`. Each case ingests a verified ledger/tool fact and a conflicting unverified chat claim in `connect_agent()`; the cue asks for the grounded value. Top-1 must be the verified engram. Domains span wallet, shipping, subscription, API quota, inventory, meeting, deployment, refund, license, and temperature. This is synthetic author-designed stress testing, not a peer benchmark, but stronger than $n=1$ for the provenance story.

Shared-brain stress is the headline (multi-tenant analogue). The thesis is provenance-weighted recall under realistic shared memory. When all 50 cases share one `connect_agent()` brain—100 conflicting engrams, semantically similar cues—top-1 collapses to **18%** (9/50; `--shared-brain`). We attribute this to *cross-case cue contamination*: unrelated domains share lexical overlap (“balance,” “quota,” “version”), so activation returns another case’s ledger or chat engram. Under *per-case isolation* (one brain per conflict pair), the same suite scores **100%** (50/50)—a ceiling under isolation, not evidence of multi-tenant readiness (Table 6). **Status:** no implementation mitigates the shared-brain failure as of this preprint—`activate_scoped` can post-filter by `agent_id`, but ingest and benchmarks do not enforce per-tenant namespaces, and we have not re-run the suite under scoped activation (Section 6).

Table 6: Provenance-conflict suite ($n=50$): shared brain is the deployment stress; isolated is the isolation ceiling.

Condition	Top-1 accuracy	Notes
Shared (one brain, all cases)	18% 9/50; cross-case contamination	
Isolated (one brain per case)	100% 50/50; isolation ceiling	

4.4 LongMemEval-S: 97.6% session recall@8 (v4 unified 500)

LongMemEval [14] tests six long-horizon abilities across 500 questions. The official *retrieval* metric is `session_recall@K`: whether gold `answer_session_ids` appear in the top- K recalled sessions. End-to-end QA—retrieve, reader LLM, GPT-4o judge—is a separate layer (Table 8).

We run retrieval with `connect_index()`, one engram per chat session, hybrid BM25 + dense recall, dual-key user indexing (CP2), heuristic query expansion (CP3), and a third *pref-facts* engram per session (v4). This hybrid-index stack does not use Recall Fabric rerank (unlike the CHORUS

benchmarks above). Embeddings use `multi-qa-mpnet-base-dot-v1` on GPU; the unified 500-question v4 run scores **97.6%** session recall@8 (488/500) and completes in ~ 73 minutes on Colab T4 (8.8s/question). The harness also supports `--report-ks 5,8,10` to score $K=5, 8$, and 10 from a single recall pass at $\max(K)=10$; frozen multi- K JSON: `longmemeval-session-multi-k-2026-07-10.json`. Reproduce: `benchmarks/longmemeval.colab-v2.ipynb` (`BENCH_PROFILE=full`) or `longmemeval_bench.py` with `--dual-key --pref-facts-key --query-expand --report-ks 5,8,10`.

Table 7: LongMemEval-S retrieval: session recall at multiple K (full 500 questions, v4 harness).

FluctlightDB (this work)	session recall	Notes
v4 @ $K=5, K=10$	<code>--report-ks 5,8,10</code> (same v4 harness; Colab ~ 75 min)	
v4 @ $K=8$	97.6%	488/500; Colab mpnet
knowledge-update	100% @8	
multi-session	98.5% @8	
single-session-user	97.1% @8	
single-session-assistant	98.2% @8	
temporal-reasoning	95.5% @8	
single-session-preference	96.7% @8	29/30
<i>Cited leaderboard figures (different K / embedders; not head-to-head)</i>		
gbrain	97.6% @5	hybrid + text-embedding-3-large
YourMemory	95.8% @5	mpnet + BM25 + graph BFS
M3 Memory	96.8% @10	FTS5 + vector + MMR

Leaderboard figures use $K=5$ or $K=10$ and different embedders; our $K=8$ run is directly comparable in *protocol* (gold session in top- K) but not a claim of strict SOTA. Preference questions rose from 76.7% (23/30) without v4 pref-facts to **96.7%** (29/30); the sole remaining preference miss is 95228167.

4.5 End-to-end QA on LongMemEval-S

We evaluate the full stack with the official LongMemEval reader prompt and Wu et al.’s task-specific GPT judge templates (`benchmarks/longmemeval_e2e.py`; `paper` profile with Muon retrieval, gold-session reader context on hard types, and type-aware readers: GPT-4o on easy types, GPT-5 on multi-session, temporal, preference, and knowledge-update). Retrieved sessions are formatted as JSON chat history; the judge is **GPT-4o** following Wu et al.’s published templates. Baselines in Table 8 use different readers and retrieval stacks; we report our numbers without claiming a like-for-like SOTA comparison.

The frozen internally reproduced run (July 2026) is `benchmarks/results/e2e-cert-paper-v2-2026-07-07.json` (487/500 correct; ~ 99 min wall time). Reproduce: `E2E_PROFILE=paper benchmarks/e2e_certify.sh`. By type, E2E accuracy is 100% on single-session user, assistant, and preference; 99.3% temporal; 97.4% knowledge-update; 92.5% multi-session—the remaining gap is reader aggregation, not retrieval. We report retrieval and QA together because a strong retrieval layer that feeds a weak reader still fails in production.

Table 8: LongMemEval-S: retrieval vs. end-to-end QA (500 questions). Vendor rows use published figures on *different* reader/retrieval stacks (not comparable to our **paper** profile).

FluctlightDB (this work)	session@8	E2E acc.	Notes
paper profile	100%	97.4%	Muon; gpt-4o/5 reader; gpt-4o judge
task-averaged E2E	—	98.2%	macro over question types
<i>Cited vendor/literature (different protocols)</i>			
Zep	—	71.2%	gpt-4o; arXiv:2501.13956
TiMem	—	76.9%	LLJ; gpt-4o-mini reader
PlugMem	—	~90%	graph memory
Mem0 (vendor)	—	94.4%	GPT-5; top-200; vendor report

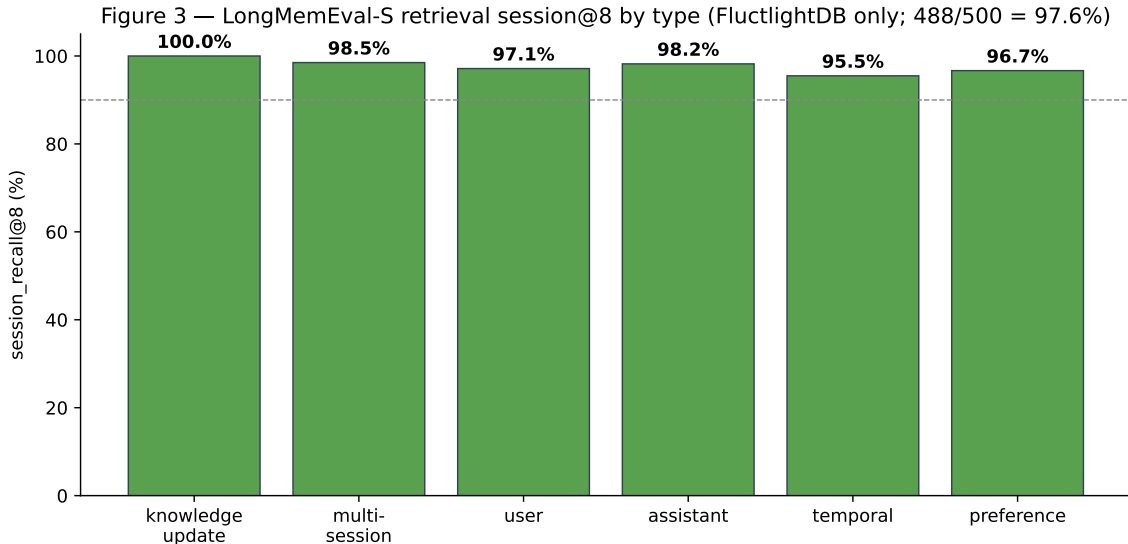


Figure 3: LongMemEval-S retrieval: session recall@8 by question type (v4 unified 488/500). Preference questions reach 96.7% (29/30) with v4 pref-facts indexing; temporal reasoning is the main remaining retrieval gap.

5 Discussion

The engine is the contribution, not the reader. A memory system should be judged on whether it surfaces the right evidence; 96.8% raw recall on full LoCoMo says it does. End-to-end QA additionally depends on the language model and the prompt budget, and we are careful not to launder a retrieval win into a generation claim.

Operational use. We have exercised FluctlightDB in long-running developer workflows (not a third-party production case study). The durability and crash-recovery properties in Section 3 are what those workflows rely on, alongside the chaos suite above—not properties asserted only from benchmark scripts.

Hybrid retrieval on measured ablations. On LoCoMo with `connect_index()` at $k=50$, hybrid BM25+dense and vector-fast-only tie within 0.1 percentage points mean evidence recall (full-set ablation JSON). The 96.8% headline uses the native CHORUS stack at $k=150$. Dialogue evidence can hinge on entities and dates; we report measured gaps rather than treating hybrid

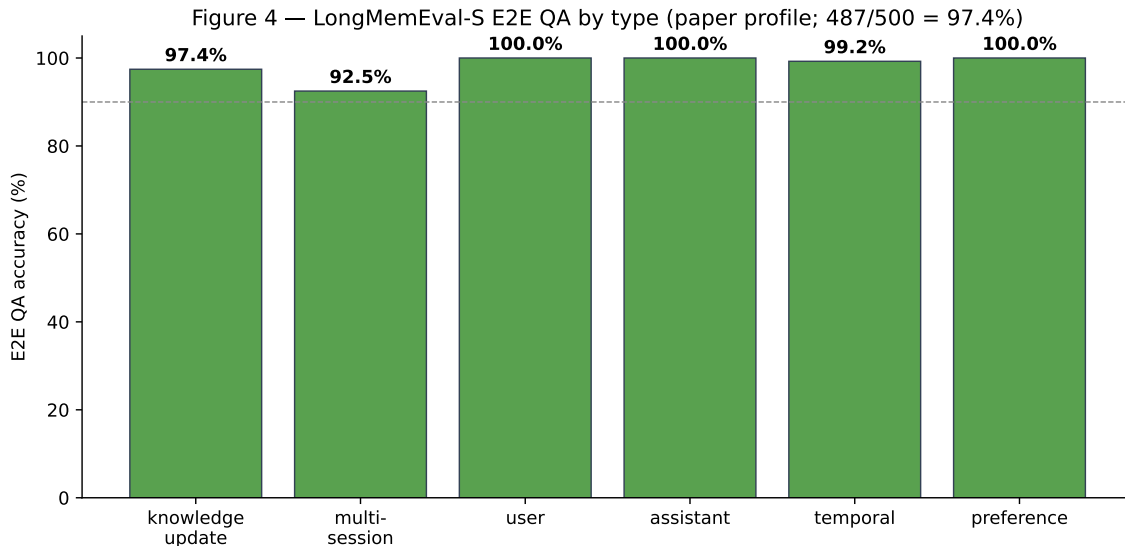


Figure 4: LongMemEval-S end-to-end QA by question type (internally reproduced **paper** profile, 487/500 overall). Readers: GPT-4o on easy types, GPT-5 on multi-session, temporal, preference, and knowledge-update; judge: GPT-4o (Wu et al. templates). Multi-session aggregation (92.5%) is the main remaining E2E gap; retrieval in the same pipeline is 100% session@8. Vendor E2E figures in Table 8 use different stacks and are not plotted here.

recall as axiomatic.

Limitations. Provenance under shared memory. The graded 50-case suite scores **18%** top-1 (9/50) when all pairs share one brain—the figure that tests the thesis under realistic co-location (Table 6). Per-case isolation reaches **100%**; we treat that only as an isolation ceiling, not multi-tenant readiness. The shared-brain drop reflects cross-case cue contamination, not a broken verified-vs-chat ranker within a single scope. **Status:** unaddressed in code and unevaluated beyond this synthetic stress test (Section 6).

LoCoMo k and metric choice. Headline LoCoMo retrieval uses $k=150$ and *raw*, unexpanded evidence recall; Table 5 reports sensitivity to smaller budgets (72.6% at $k=5$, MiniLM). The deprecated 99% figure relied on ± 3 neighbor expansion and is not used anywhere in this paper.

LongMemEval K . We report session recall at $K=8$ from a frozen Colab run and document a `--report-ks 5,8,10` harness for $K=5$ and $K=10$ from one recall pass; leaderboard rows still mix embedders and protocols.

LoCoMo end-to-end gap. Full LoCoMo retrieval reaches **96.8%** raw evidence recall (MiniLM; 97.0% mpnet), and an end-to-end QA read over *date-stamped* context reaches $\approx$ **85%** accuracy at $k=15$. Because gold LoCoMo answers are usually inferred facts (dates, counts) rather than quoted spans, the reader must see *when* each turn happened; without date-stamped context, temporal questions fail and E2E accuracy collapses far below retrieval. The remaining retrieval→QA gap is therefore reader/formatting bound, not a retrieval failure. Evidence-recall and LLM-judge QA are reported as separate metrics and never conflated.

Retrieval-only v4 (Figure 3) still has temporal and preference misses at 95.5% and 96.7%; LLM-based key expansion (LongMemEval CP2 full) is not yet in the default ingest path. End-to-end QA on LongMemEval-S (Table 8, Figure 4) uses OpenAI reader/judge APIs and remains weakest on multi-session aggregation (92.5%). Mem0/Zep end-to-end figures use different reader setups; we cite published LongMemEval-S numbers only where protocols match. ANN beyond 10^5 memories

is not yet evaluated.

6 Future Work

The shared-brain provenance result (**18%** top-1; 100% only under isolation) is the highest-priority gap for a revised preprint: session- or agent-scoped **experience/activate**, stricter hybrid candidate filtering, and a re-benchmark of `provenance_conflict_bench.py` under realistic multi-tenant load. That work is **not implemented** in the repository accompanying this submission; a future version should report before/after top-1 accuracy in Table 6. Secondary targets include full Long-MemEval session recall at $K=5$ and $K=10$ from the `--report-ks` harness, head-to-head LoCoMo evidence recall against published Mem0/Zep figures where protocols align, and replacing the hybrid SQLite sidecar with a native in-engine recall index.

7 Related Work

7.1 Vector databases

Systems such as Chroma, Qdrant, and FAISS [12] optimize approximate nearest-neighbor search. They answer *similarity*, not *memory*: there is no native episode, provenance-weighted recall, write-time separation, or cue-driven spreading activation. FluctlightDB’s `connect_index()` mode deliberately exposes a vector-fast path and a CHORUS bulk-imprint lane so we can measure IR credibility on BEIR; episodic agent mode adds graph co-activation for live agents.

7.2 Agent memory SDKs and layers

Mem0 [1] is the closest contemporary system: it extracts, consolidates, and retrieves salient facts from dialogue, with an optional graph variant (Mem0^g) for relational structure among entities. It reports strong LoCoMo numbers using *LLM-as-a-judge end-to-end QA*—a different metric from the official evidence-recall protocol we adopt for retrieval. **Zep** [15] provides temporal knowledge-graph memory for assistants, also evaluated with reader-LLM QA on LoCoMo. **Cognee** [13] pipelines vector and graph storage for agent RAG. **MemGPT** [9] (Letta) treats context as an OS resource with explicit memory tiers but does not define a standalone embedded memory *engine contract*. **HippoRAG** [4] draws on associative-retrieval neuroscience for multi-hop QA over a graph index. These systems are valuable *memory layers*: they orchestrate extraction, summarization, and retrieval over general-purpose backends. We position FluctlightDB as the missing *engine* beneath such layers—where the native operations are **experience** and **activate**, not SQL or **vector.search**.

7.3 Positioning

We propose an engine-level data model for long-term agent memory—episodes, cues, provenance on write; activation on read—distinct from SQL rows and vector ANN. We make this case carefully, not categorically: Mem0 [1], Mem0^g, Zep [15], and HippoRAG [4] already treat memory as structured facts, graphs, or associative indices with provenance-like trust policies in application layers. Our claim is narrower: an **embedded** store whose native contract is **experience/activate**, with durability, separation, and hybrid recall in one binary—a substrate those layers can sit on, not a replacement for their extraction and orchestration. Enterprise databases adding agent features (e.g. Oracle AI Database unified memory) retrofit memory onto existing engines rather than defining

episodes and cues as first-class types. Table 9 summarizes the landscape; direct head-to-head retrieval on LoCoMo evidence recall against Mem0/Zep is future work (different published metrics today).

Table 9: Landscape: memory *layers* vs. FluctlightDB as an embedded *engine*. **Top block:** our internally reproduced metrics. **Bottom block:** cited vendor/literature figures on *different* metric types—not head-to-head.

System	Kind	Native tract	con-	Benchmark	Metric type
<i>FluctlightDB (this work; internally reproduced)</i>					
FluctlightDB	Engine	experience activate	/	96.8%	LoCoMo evidence (raw, $k=150$)
FluctlightDB	Engine	hybrid index + session keys	+	97.6%	LongMemEval session@8
FluctlightDB	Engine	Muon + E2E	paper	97.4%	LongMemEval E2E QA
<i>Cited vendor/literature (different metrics; not comparable)</i>					
Mem0 Mem0 ^g	/ SDK backend	+ Extract / consolidate / retrieve facts	/ con- / re-	~92%+ LLM-J	End-to-end QA
Zep	Managed layer	Temporal + summaries	KG	~75% LLM-J	End-to-end QA
Cognee	Pipeline	Graph + vector ETL		—	Task-specific
MemGPT Letta	/ Agent OS	Context tiers / blocks	/	—	Session QA
HippoRAG	Graph RAG	Associative graph retrieval		—	Multi-hop QA
Chroma Qdrant	/ Vector DB	ANN top- k		—	IR / similarity

Benchmarks LoCoMo [8] and LongMemEval [14] measure long-horizon memory; BEIR [12] anchors IR credibility. Our framing—memory as a *proposed* data model on equal footing with rows and vectors—is the argument this engine and its measurements are built to test.

8 Conclusion

The relational model gave applications a database for *facts*; the vector model gave search a database for *similarity*. Autonomous agents need a database for *memory*, and it should be as boring to adopt and as rigorous to trust as SQLite. FluctlightDB is our argument that this engine can exist today: it ties vector baselines where they are strong on shared harnesses, reaches 96.8% raw LoCoMo evidence *retrieval* (no neighbor expansion) with $\approx 85\%$ end-to-end QA over date-stamped context, scores 97.6% session recall and 97.4% end-to-end QA on LongMemEval-S, and—on the provenance thesis under shared memory—only 18% top-1 on the graded conflict suite (100% under isolation). Harnesses, frozen JSON, and MIT engine source are released so others can re-run and contest these claims; the one-minute path uses the published wheel.

A Recall Fabric mechanisms

Hybrid recall (Section 3) treats memory as an index. FluctlightDB also implements a *Recall Fabric*—Photon prefilter (SimHash/LSH), Manifold Lattice addressing [3, 11], theta–gamma phase parsing [6, 7, 10], consolidation crystallization, adaptive forgetting, chronos time buckets, and confidence-weighted consensus—each as a deterministic, dependency-free module behind `FLUCTLIGHT_FABRIC=1`. On ingest, Fabric indexes engrams (and CHORUS traces) for lattice/phase scoring; on recall it re-ranks hybrid and CHORUS shortlists before returning hits. Synthetic property tests cover CRT round-trip, bind/unbind invertibility, Hamming–cosine rank agreement, and decay monotonicity. Unlike the headline results in Section 4, these mechanisms are *not* validated as standalone peer benchmarks—only as optional rerank on CHORUS shortlists where Fabric was enabled. Extended mechanism notes live in repository documentation.

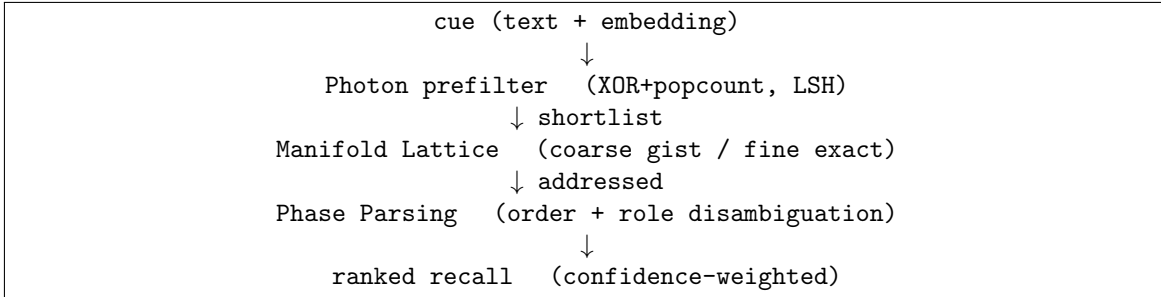


Figure 5: Recall Fabric pipeline (`FLUCTLIGHT_FABRIC=1`). Composes with CHORUS recall on LoCoMo, BEIR, and FAMB frozen runs (Section 4).

Artifacts

Repository: FluctlightDB (MIT). Harnesses: `benchmarks/locomo_eval.py`, `benchmarks/locomo_ablation.py`, `benchmarks/provenance_conflict_bench.py`, `benchmarks/longmemeval_bench.py`, `benchmarks/longmemeval_e2e.py`, `benchmarks/beir_bench.py`, `benchmarks/agent_memory_bench.py`. Frozen metrics: `benchmarks/results/paper`. E2E: `benchmarks/longmemeval_e2e.py`; `benchmarks/results/e2e-cert-paper-v2-2026-07-07.json`. Preprint DOI: <https://doi.org/10.5281/zenodo.20949890>. Source: <https://github.com/voxmastery/FluctlightDB/tree/main/papers/arxiv-v1>.

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